Rishabh Jain

Senior Software Engineer | Distributed Systems, Applied AI and Data Science

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PROFESSIONAL EXPERIENCE

Microsoft
2024 to present

Senior Software Engineer | Azure SQL Database

- Improved CPU allocation through differential overbooking and rebalancing changes, supporting an overbooking increase that cut the physical CPU capacity needed for the same workloads by about 4.6%.
- Wrote the CPU-rebalancing technical vision, defined effectiveness metrics and dashboards, and reviewed the resulting implementations. Started and led a recurring design discussion series.
- Implemented hardware-aware (NUMA) CPU allocation for containerised workloads.
- Designed failover diagnostics and automated alerting, coordinating reviews with three partner teams.
- Built an AI investigation agent, now used by the team, with 30+ skills for incident analysis, operational reporting and deployment correlation, grounded in 800+ existing diagnostic queries.
- Built an AI-assisted code-analysis tool that surfaced about 10 latent defects in resource-governance code, and delivered fixes for all of them.
- Diagnosed production incidents as an expert on-call engineer, using telemetry and memory dumps.

Technologies: C++, Azure SQL, Kusto (KQL) and WinDbg

Google
2021 to 2024

Software Engineer III (L4) | Encrypted Editors: Docs, Sheets and Slides

- Led a seven-engineer team building encrypted import and export infrastructure. Improved chunked-import fidelity from 93.3% to 100% on an internal corpus of real-world files.
- Designed and implemented Excel (.xlsx and .xls) imports for encrypted Sheets and Word (.docx) imports for encrypted Docs using Web Workers.
- Implemented Excel exports of encrypted documents for Google Takeout, achieving over 99.99% fidelity for encrypted Sheets in a test corpus.
- Profiled client-side import and export, and brought their performance to parity with server-side processing.
- Conducted 50+ technical interviews, managed an intern and led Women Engineers workshops.

Technologies: Java, Web Workers and Spanner

Microsoft
2018 to 2021

Software Engineer II | Microsoft Teams

- Scaled Redis caching without downtime for notification services processing 2.1 billion events daily.
- Reduced partner network calls by 60%. Extended cache lifetimes with more precise invalidation, consolidated APIs and reordered requests to avoid unnecessary downstream calls.
- Added customer-level encryption in Azure Storage Queues and improved resilience through stale caching and data-driven retry policies.
- Supported business continuity and disaster recovery testing and monitoring dashboards.
- Analysed notification-email click-through rates to identify the products where Teams was used most.

Technologies: .NET Framework, .NET Core, Azure Storage and Redis

Kuliza | Software Developer Intern
2018

Developed asynchronous caching, role-based access controls, and lead generation and aggregation for HDFC's brand website and CMS administration.

HackerRank | Software Engineering Intern
June to July 2017

Created SQL, algorithm and regular-expression challenges for hiring assessments.

TECHNICAL SKILLS

Languages: C++, Java, C#, Python, SQL and JavaScript

Backend and cloud: .NET Framework, .NET Core, Guice, Azure Cloud Services and Service Fabric

Data and messaging: Azure SQL, Spanner, Cosmos DB, Redis, Azure Tables, Blob Storage and Storage Queues

Analysis and ML: Kusto (KQL), pandas, NumPy, GeoPandas, LightGBM, Keras and TensorFlow

Development and diagnostics: Git, Docker, WinDbg, Jupyter, Matplotlib and Seaborn

PROJECT WORK

Legal Document RAG (LegalBench-RAG 2024) 2026

Python, Chroma, OpenAI and Ragas | Report | Repository

- Built contract-scoped retrieval over 644 contracts and 63,914 chunks, with cited answers and abstention. Retrieved reference evidence for nine of ten sampled benchmark questions (Hit@5: 0.90).

How Delhi Buses Move 2026

Python, GTFS-Realtime, pandas and LightGBM | Report | Repository

- Processed six hours of GPS from 6,065 buses for trajectory analysis and arrival modelling. Reduced arrival-time mean absolute error by 14% against a historical-median baseline in a time-matched evaluation.

Demand-Supply Workforce Heatmaps 2019; revisited in 2026

Android, Django, SQLite and Keras (LSTM experiment) | Report | Repository

- Built a team prototype comparing synthetic demand and resources across ten Bengaluru zones. Placed third nationally at Zomato Hackanoodle and presented to Deepinder Goyal. Added regression tests and an interactive report in 2026.

NYC Yellow Taxi Analysis 2026

Python, pandas and GeoPandas | Report | Repository

- Analysed a 0.75% sample from 38.3 million 2023 trip records to examine demand, geography and fares. Published an interactive report with data-quality checks and limitations on recorded-tip analysis.

EDUCATION

MS in Machine Learning and Artificial Intelligence 2025 to present

Liverpool John Moores University, via UpGrad. Includes an Executive Diploma in Machine Learning and AI from IIIT Bangalore.

B.Tech in Computer Science and Engineering 2014 to 2018

Jaypee Institute of Information Technology, Noida. Honours; CGPA: 8.9; Best Outgoing Student

MENTORING

- Mentored 15+ learners over 100+ hours through Scaler Academy, 10xAcademy and other programmes, covering system design, backend development and interview preparation.
- Conducted mock interviews and supported mentees who secured roles at organisations including Amazon, Google and Microsoft.

AWARDS AND RECOGNITION

- Received the Google Cloud Tech Impact Award in 2022 for client-side encryption work.
- Received three Google Grungy Awards for addressing technical debt, infrastructure issues and reliability.
- Represented JIIT at ICPC Amritapuri in 2017 and achieved 97th All India Rank.